

Mathematical Foundations of Political Methodology

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What is this half of the course about?

- The science of describing uncertainty
- How to separate the signal from the noise
- More problem solving with computers

Origins of Probability Theory

In 1717, gamblers played a dice game in which you would make the following bet:

Bet A At least one  will appear in a total of four rolls of a die.

- *Antoine Gombaud, Chevalier de Méré* proposed a tougher game, rolling *two* die, looking for a pair ().
- He figured, for a single trial, the chance of getting  was one 1/6th as likely as  :

$$\rightarrow P(\text{) = P(\text{$$

Bet B At least one  will appear in a total of 24 rolls of two dice.

Origins of Probability Theory

- Over the course of his holiday, the Chevalier de Méré tried his bets on his friends
- Bet A worked quite well, Antoine would get a  in four rolls of a die more often than not.
- Bet B was working worse. His friends were winning more than he was losing.

```
> 1:6 # Sample space of a single die  
[1] 1 2 3 4 5 6
```

```
> sample(1:6, size = 4, replace=TRUE)  
[1] 4 4 2 2
```

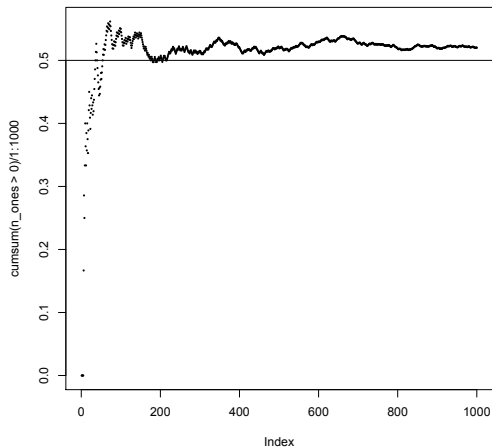
```
> onedice4x <- replicate( n = 1000,  
sample(1:6, size = 4, replace=TRUE))
```

```
> n_ones <- colSums( onedice4x == 1)
```

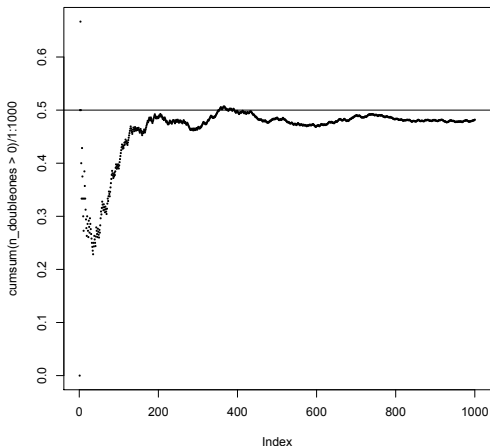
```
> table(n_ones>0) # Bet A outcomes:
```

```
FALSE  TRUE  
  480   520
```

```
> plot(cumsum(n_ones>0)/1:1000, cex=.2)  
> abline(h=.5)
```



```
> twodice <- outer(1:6, 1:6, paste) # Outcomes for two dice  
> twodice24x <- replicate( n = 1000,  
  sample(twodice, size = 24, replace=TRUE))  
> n_doubleones <- colSums(twodice24x=="1 1")
```



Where did Antoine go wrong?

- $P(\begin{array}{|c|c|}\hline \bullet & \bullet \\ \hline\end{array})$ is $1/6$ as likely as $P(\begin{array}{|c|}\hline \bullet \\ \hline\end{array})$:
 - One roll, one die = $\{\begin{array}{|c|}\hline \bullet \\ \hline\end{array}, \begin{array}{|c|}\hline \bullet \bullet \\ \hline\end{array}, \begin{array}{|c|}\hline \bullet \bullet \bullet \\ \hline\end{array}, \begin{array}{|c|}\hline \bullet \bullet \bullet \bullet \\ \hline\end{array}, \begin{array}{|c|}\hline \bullet \bullet \bullet \bullet \bullet \\ \hline\end{array}, \begin{array}{|c|}\hline \bullet \bullet \bullet \bullet \bullet \bullet \\ \hline\end{array}\} \rightarrow 1/6$
 - One roll, two die = $\{(\begin{array}{|c|}\hline \bullet \\ \hline\end{array}, \begin{array}{|c|}\hline \bullet \\ \hline\end{array}), (\begin{array}{|c|}\hline \bullet \\ \hline\end{array}, \begin{array}{|c|}\hline \bullet \bullet \\ \hline\end{array}), (\begin{array}{|c|}\hline \bullet \\ \hline\end{array}, \begin{array}{|c|}\hline \bullet \bullet \bullet \\ \hline\end{array}), (\begin{array}{|c|}\hline \bullet \\ \hline\end{array}, \begin{array}{|c|}\hline \bullet \bullet \bullet \bullet \\ \hline\end{array}), (\begin{array}{|c|}\hline \bullet \\ \hline\end{array}, \begin{array}{|c|}\hline \bullet \bullet \bullet \bullet \bullet \\ \hline\end{array}), (\begin{array}{|c|}\hline \bullet \\ \hline\end{array}, \begin{array}{|c|}\hline \bullet \bullet \bullet \bullet \bullet \bullet \\ \hline\end{array}), (\begin{array}{|c|}\hline \bullet \bullet \\ \hline\end{array}, \begin{array}{|c|}\hline \bullet \\ \hline\end{array}), (\begin{array}{|c|}\hline \bullet \bullet \\ \hline\end{array}, \begin{array}{|c|}\hline \bullet \bullet \\ \hline\end{array}), (\begin{array}{|c|}\hline \bullet \bullet \\ \hline\end{array}, \begin{array}{|c|}\hline \bullet \bullet \bullet \\ \hline\end{array}), (\begin{array}{|c|}\hline \bullet \bullet \\ \hline\end{array}, \begin{array}{|c|}\hline \bullet \bullet \bullet \bullet \\ \hline\end{array}), (\begin{array}{|c|}\hline \bullet \bullet \\ \hline\end{array}, \begin{array}{|c|}\hline \bullet \bullet \bullet \bullet \bullet \\ \hline\end{array}), (\begin{array}{|c|}\hline \bullet \bullet \\ \hline\end{array}, \begin{array}{|c|}\hline \bullet \bullet \bullet \bullet \bullet \bullet \\ \hline\end{array}), (\begin{array}{|c|}\hline \bullet \bullet \bullet \\ \hline\end{array}, \begin{array}{|c|}\hline \bullet \\ \hline\end{array}), (\begin{array}{|c|}\hline \bullet \bullet \bullet \\ \hline\end{array}, \begin{array}{|c|}\hline \bullet \bullet \\ \hline\end{array}), (\begin{array}{|c|}\hline \bullet \bullet \bullet \\ \hline\end{array}, \begin{array}{|c|}\hline \bullet \bullet \bullet \\ \hline\end{array}), (\begin{array}{|c|}\hline \bullet \bullet \bullet \\ \hline\end{array}, \begin{array}{|c|}\hline \bullet \bullet \bullet \bullet \\ \hline\end{array}), (\begin{array}{|c|}\hline \bullet \bullet \bullet \\ \hline\end{array}, \begin{array}{|c|}\hline \bullet \bullet \bullet \bullet \bullet \\ \hline\end{array}), (\begin{array}{|c|}\hline \bullet \bullet \bullet \\ \hline\end{array}, \begin{array}{|c|}\hline \bullet \bullet \bullet \bullet \bullet \bullet \\ \hline\end{array}), (\begin{array}{|c|}\hline \bullet \bullet \bullet \bullet \\ \hline\end{array}, \begin{array}{|c|}\hline \bullet \\ \hline\end{array}), (\begin{array}{|c|}\hline \bullet \bullet \bullet \bullet \\ \hline\end{array}, \begin{array}{|c|}\hline \bullet \bullet \\ \hline\end{array}), (\begin{array}{|c|}\hline \bullet \bullet \bullet \bullet \\ \hline\end{array}, \begin{array}{|c|}\hline \bullet \bullet \bullet \\ \hline\end{array}), (\begin{array}{|c|}\hline \bullet \bullet \bullet \bullet \\ \hline\end{array}, \begin{array}{|c|}\hline \bullet \bullet \bullet \bullet \\ \hline\end{array}), (\begin{array}{|c|}\hline \bullet \bullet \bullet \bullet \\ \hline\end{array}, \begin{array}{|c|}\hline \bullet \bullet \bullet \bullet \bullet \\ \hline\end{array}), (\begin{array}{|c|}\hline \bullet \bullet \bullet \bullet \\ \hline\end{array}, \begin{array}{|c|}\hline \bullet \bullet \bullet \bullet \bullet \bullet \\ \hline\end{array})\} \rightarrow 1/36$
 - Using these enumerated possibilities, we can calculate probabilities.
- Much of probability is about reasoning backward.

Solving by enumeration

Roll a die 4 times, the total number of possible outcomes:

$$6 \times 6 \times 6 \times 6 = 1296$$

Of these, there are 625 outcomes **without** a :

$$5 \times 5 \times 5 \times 5 = 625$$

So, Antoine wins bet A

$$P(A) = \frac{1296 - 624}{1296} \approx 0.52$$

Using Probabilistic Reasoning

So how about when there are 24 rolls?

$$P(\text{no } \blacksquare \text{ in 4 rolls}) = (5/6) * (5/6) * (5/6) * (5/6) = (5/6)^4$$

$$P(A) = 1 - (5/6)^4 \approx .52$$

$$P(\text{no } \blacksquare \blacksquare \text{ in 24 rolls}) = (35/36)^{24}$$

$$P(B) = 1 - (35/36)^{24} \approx .49$$

Sets and Probability

Probability model represent random phenomenon using mathematical set notation:

Sample Spaces: All Things that Can Happen

Definition

The *sample space* is the set of all things that can occur. This set is often referred to by the symbol Ω or S .

Examples:

1) The outcome of two dice

$$\Omega = \left\{ \begin{array}{|c|c|} \hline \bullet & \bullet \\ \hline \end{array}, \begin{array}{|c|c|} \hline \bullet & \bullet \\ \hline \end{array}, \dots, \begin{array}{|c|c|} \hline \bullet & \bullet \\ \hline \end{array} \right\}$$

2) Survival of Monarch

$$\Omega = \{t : 0 \leq t \leq 120\}$$

Events: Subsets of Sample Space

Events are **sets**: collection of distinct objects.

$$A = \{(Sanders, Ellison), (McMullin, Pence)\}$$

$$B = \{(Trump, Pence), (Trump, Duke), (McMullin, Pence)\}$$

$$\Omega = \{(Trump, Pence), (Sanders, Ellison), (Trump, Duke), (McMullin, Pence), (Booker, Harris)\}$$

$$A \subset \Omega, B \subset \Omega, \Omega \subseteq \Omega$$

We can make a new set E , using *set operations*

1) Union: $\cup$

- All objects that appear in **either** set
- $E = A \cup B = \{(Trump, Pence), (Sanders, Ellison), (Trump, Duke), (McMullin, Pence)\}$

2) Intersection: $\cap$

- All objects that appear in **both** sets
- $F = A \cap B = \{(McMullin, Pence)\}$
- Sometimes written as AB

Events: Subsets of Sample Space

Definition

Suppose E and F are events. If $E \cap F = \emptyset$ then we'll say E and F are *mutually exclusive*

- Mutual exclusive events cannot mutually occur
- E and E^c are mutually exclusive events
- If Trump wins reelection, Booker will not have been elected president.
- If E is mutually exclusive of F , If I know that E happened, I can be sure that F did not.

Events: Subsets of the Sample Space

Definition

Suppose we have events $E_1, E_2, \dots, E_N$.

Define:

$$\cup_{i=1}^N E_i = E_1 \cup E_2 \cup E_3 \cup \dots \cup E_N$$

$\cup_{i=1}^N E_i$ is the set of outcomes that occur at least once in $E_1, \dots, E_N$. This is like a logical or.

Define:

$$\cap_{i=1}^N E_i = E_1 \cap E_2 \cap \dots \cap E_N$$

$\cap_{i=1}^N E_i$ is the set of outcomes that occur in each E_i . This is like a logical and.

Conditions for things to be events

Condition

Ω must be an event.

We aren't leaving out Ω .

Condition

If A is an event, A^c is an event.

We can't leave the sample space by taking complements

Condition

If $A_1, A_2, A_3 \dots$ are a countable collection of events, then $\bigcup_{i=1}^{\infty} A_i$ is an event.

We cannot leave out combinations of events.

Probability Law

- **Probability:** chance of event
 - P is a function $\Omega \rightarrow [0, 1]$
 - Describes relative likelihood of all E .

Probability Axioms

Definition

All probability functions, P , satisfy three axioms:

- 1) (Nonnegativity) *For every event A ,*

$$P(A) \geq 0$$

Chances are never negative

- 2) (Additivity) *For all sequences of mutually exclusive events $E_1, E_2, \dots, E_N$ (where N can go to infinity)*

$$P\left(\bigcup_{i=1}^N E_i\right) = \sum_{i=1}^N P(E_i)$$

- 3) (Normalization) *$P(\Omega) = 1$ The probability of any certain events are 1*

Probability Examples

- Suppose we are flipping a **fair** coin. Then $P(\{H\}) = P(\{T\}) = 1/2$
- Suppose we are rolling a six-sided die. Then $P(\{1\}) = 1/6$
- Suppose we are flipping a pair of fair coins. Then $P(\{(H, H)\}) = 1/4$

Properties of Probability

- If $A \subseteq B$, $P(A) \leq P(B)$
- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- $P(A \cup A^c) = 1$
- $P(A \cap A^c) = 0$
- $P(\emptyset) = 0$
- $P(E) = 1 - P(E^c)$

Properties of Probability: Monotonicity

Proposition

If $E \subset F$ then $P(E) \leq P(F)$.

Proof.

We can write $F = E \cup (E^c \cap F)$. (Why?)

Further, $E \cap (E^c \cap F) = \emptyset$

Then

$$P(F) = P(E) + P(E^c \cap F)$$



As you add more “outcomes” to a set, it can’t reduce the probability.

Properties of Probability

Proposition

Consider events E_1 and E_2 . Then

$$P(E_1 \cap E_2) = P(E_1) - P(E_1 \cap E_2^c)$$

Proof.

$$\begin{aligned} E_1 &= (E_1 \cap E_2) \cup (E_1 \cap E_2^c) \\ P(E_1) &= P(E_1 \cap E_2) + P(E_1 \cap E_2^c) \\ P(E_1 \cap E_2) &= P(E_1) - P(E_1 \cap E_2^c) \end{aligned}$$



Counting and the binomial formula

If we have m elements, $\binom{m}{k}$ is number of ways to choose a subset of size k .

If we have m sequences of binary outcomes, $\binom{m}{k}$ is number of sequences that contain k successes.

$$\binom{m}{k} \equiv \frac{m!}{k!(m-k)!}$$

$$i! \equiv 1 \cdot 2 \cdots (i-1) \cdot i$$

Binomial Theorem:

$$(x + y)^2 = x^2 + 2xy + y^2$$

$$(x + y)^3 = x^3 + 3x^2y + 3xy^2 + y^3$$

$$(x + y)^4 = x^4 + 4x^3y + 6x^2y^2 + 4xy^3 + y^4$$

...

$$(x + y)^n = \sum_{i=0}^n \binom{n}{i} x^{n-i} y^i$$

Examples of counting

- How many different combinations can occur when we throw two dice that aren't pairs?
- The number of elements $m = 6$, the subset is size $k = 2$

$$\begin{aligned}\binom{m}{k} &= \frac{6!}{2!(6-2)!} \\ &= \frac{6 \times 5 \times 4 \times 3 \times 2}{2 \times 4 \times 3 \times 2} \\ &= \frac{6 \times 5}{2} = 15\end{aligned}$$

■

Examples of counting

- How many different networks can occur if there are 24 people?
- A network is characterized by the presence or absence of a connection (edge) between people (nodes).
- If there are m nodes, there are $\binom{m}{2} = \frac{m!}{2!(m-2)!} = m(m-1)/2$ edges in a complete graph.
- Each of these edges may be present or absent, so there are two possibilities for each edge, so in if $m = 6$:

$$2 * 2 * 2 * 2 * 2 * 2 * 2 * 2 * 2 * 2 * 2 * 2 * 2 * 2 * 2 * 2 = 2^{15} = 32,768$$

- If $m = 24$, the possibilities are $2^{\frac{24*23}{2}} = 2^{276}$
- For comparison, there are 2^{265} atoms in the universe.

Examples of counting

- How many observations are there in the MID's dataset?
- There are about 180 countries in the world.
- The number of undirected dyads is therefore $\binom{180}{2} = 16,110$
- The number of directed dyads (where order matters) is twice this.
- How about when you have four-digit trade data?
 $\binom{180 \times 9999}{2} = 1,600,000,000,000$

Example of binomial

- Suppose we shuffle a 52 card deck and begin flipping over the cards. What is the probability that the 13 card is the first King?
- If there is exactly one King in the first thirteen cards, the probability that it is number 13 is $1/13$.
- So what are the chances of assembling 13 cards with one king in them? Lets compare the total possibilities to the number of successes.
 - How many total ways are there of getting 13 cards out of 52? $\binom{52}{13}$
 - There are 4 kings, so there are four ways to get that 13th card to be a king.
 - How about the 12 cards that are not kings? $\binom{48}{12}$
- So the probability is $\frac{1}{13} \cdot \frac{4 \cdot \binom{48}{12}}{\binom{52}{13}}$

```
deck <- outer( c("A", 2:10, "J", "Q", "K"),
              c("h", "d", "s", "c"), paste0)
Kings <- paste0("K",  c("h", "d", "s", "c"))

trial <- function(){
first13 <- sample(as.vector(deck), 13)
Kingisthere <- first13[13] %in% Kings
KingisAlone <- !(TRUE %in% (first13[1:12] %in% Kings))
return(Kingisthere & KingisAlone)
}

out <- replicate(500000, trial())

mean(out)
[1] 0.033804

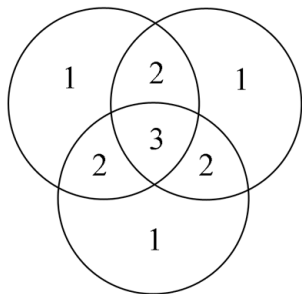
(1/13)*(4*choose(48,12))/choose(52, 13)
[1] 0.0337575
```

Inclusion/Exclusion

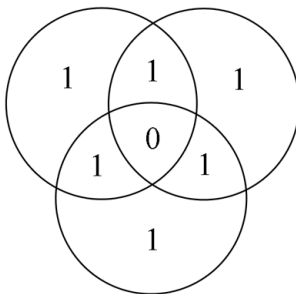
Proposition

Suppose $E_1, E_2, \dots, E_n$ are events. Then

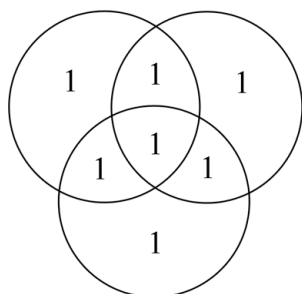
$$\begin{aligned} P(E_1 \cup E_2 \cup \dots \cup E_n) &= \sum_{i=1}^n P(E_i) - \sum_{i_1 < i_2} P(E_{i_1} \cap E_{i_2}) + \dots \\ &\quad + (-1)^{r+1} \sum_{i_1 < i_2 < \dots < i_r} P(E_{i_1} \cap E_{i_2} \cap \dots \cap E_{i_r}) \\ &\quad + \dots + (-1)^{n+1} P(E_1 \cap E_2 \cap \dots \cap E_n) \end{aligned}$$



$$|A| + |B| + |C|$$



$$|A| + |B| + |C| - (|A \cap B| + |A \cap C| + |B \cap C|)$$



$$|A| + |B| + |C| - (|A \cap B| + |A \cap C| + |B \cap C|) + |A \cap B \cap C|$$

Proof of Inclusion Exclusion

- Suppose that we have an outcome.
- If it isn't in the event sequence, doesn't appear anywhere.
- If it is in the event sequence, appears once in $\cup_{i=1}^n E_i$, it is in one portion of the Venn diagram, and contributes once to $P(\cup_{i=1}^n E_i)$.
- So how do we characterize the addition and subtraction of sets to make it count once? Suppose it appears in m of the E_i $m > 0$

$$\text{count} = \binom{m}{1} - \binom{m}{2} + \binom{m}{3} - \cdots + (-1)^{m+1} \binom{m}{m}$$

$$\text{count} = \sum_{i=1}^m \binom{m}{i} (-1)^{i+1}$$

$$\text{count} = - \sum_{i=1}^m \binom{m}{i} (-1)^i$$

Proof of Inclusion Exclusion

$$\text{count} = - \sum_{i=1}^m \binom{m}{i} (-1)^i$$

Binomial Theorem: $(x + y)^n = \sum_{i=0}^n \binom{n}{i} (x)^{n-i} y^i$.

$$0 = (1 - 1)^m = \sum_{i=0}^m \binom{m}{i} (1)^{m-i} (-1)^i$$

$$0 = \binom{m}{0} (-1)^0 + \sum_{i=1}^m \binom{m}{i} (-1)^i$$

$$0 = 1 + \sum_{i=1}^m \binom{m}{i} (-1)^i$$

$$0 = 1 - \text{count}$$

$$1 = \text{count}$$

Inclusion/Exclusion

Corollary

Suppose E_1 and E_2 are events. Then

$$P(E_1 \cup E_2) = P(E_1) + P(E_2) - P(E_1 \cap E_2)$$

Relation to Gamma Function

$$\binom{m}{k} = \frac{m!}{k!(m-k)!} = \frac{\Gamma(m+1)}{\Gamma(k+1)\Gamma(m-k+1)}$$

If n is an integer:

$$\Gamma(n) = (n-1)!$$

but if z is some real (or complex) number which is not a negative integer:

$$\Gamma(z) = \int_0^{\infty} x^{z-1} e^{-x} dx$$

$$\Gamma(1) = 1$$

$$\Gamma(1/2) = \sqrt{\pi}$$

Why are Gamma Functions useful?

- Gamma functions show up in popular distributions: exponential distribution, chi-squared distribution, F distribution, and the Student's t distribution.
- They are fundamental to many actual processes.

Proposition

Boole's Inequality

$$P(\cup_{i=1}^N E_i) \leq \sum_{i=1}^N P(E_i)$$

Proof.

Proceed by induction. Trivially true for $n = 1$. Now assume the proposition is true for $n = k$ and consider $n = k + 1$.

$$P(\cup_{i=1}^k E_i \cup E_{k+1}) = P(\cup_{i=1}^k E_i) + P(E_{k+1}) - P(\cup_{i=1}^k E_i \cap E_{k+1})$$

Proof Continued

$$P(\cup_{i=1}^k E_i) \leq \sum_{i=1}^k P(E_i)$$

$$P(\cup_{i=1}^k E_i) + P(E_{k+1}) - P(\cup_{i=1}^k E_i \cap E_{k+1}) \leq \sum_{i=1}^k P(E_i) + P(E_{k+1})$$

$$P(\cup_{i=1}^k E_i \cup E_{k+1}) \leq \sum_{i=1}^{k+1} P(E_i)$$

$$P(\cup_{i=1}^{k+1} E_i) \leq \sum_{i=1}^{k+1} P(E_i)$$

Proposition

Bonferroni's Inequality

$$P(\cap_{i=1}^n E_i) \geq 1 - \sum_{i=1}^n P(E_i^c)$$

Proof.

$\cup_{i=1}^n E_i^c = (\cap_{i=1}^n E_i)^c$. So,

$$P(\cup_{i=1}^n E_i^c) \leq \sum_{i=1}^n P(E_i^c)$$

$$\begin{aligned} P(\cup_{i=1}^n E_i^c) &= P((\cap_{i=1}^n E_i)^c) \\ &= 1 - P(\cap_{i=1}^n E_i) \end{aligned}$$

$$P(\cap_{i=1}^n E_i) \geq 1 - \sum_{i=1}^n P(E_i^c)$$



Applications of Bonferroni's Inequality

- Suppose that I learned that 10 events occur with probability .98 and know nothing else

Bonferroni lower bound:

$$P(A_1 \cap A_2 \dots \cap A_{10}) \geq 1 - .02 * 10 = .8$$

- “Bonferroni correction”: if you have m hypothesis tests with significance level $\alpha = .05$, each should be evaluated at α/m

Odds of Events

Definition

Define the odds of some event E as

$$\text{odds}_E = \frac{P(E)}{1 - P(E)}$$

Suppose F is another event. Define the **odds ratio** of E to F as

$$\begin{aligned}\text{odds ratio}_{E:F} &= \frac{\text{odds}_E}{\text{odds}_F} \\ &= \frac{\frac{P(E)}{1 - P(E)}}{\frac{P(F)}{1 - P(F)}}\end{aligned}$$

- Big: implies E is very likely
- Small: implies E is unlikely

Uncountable Sample Space

Probability of events on the real line are only positive for intervals.

- Let $E = \{x : x \in (a, b)\}$

$$P(E) = \int_{x \in E} f_X(x) dx$$

Note: Zero Probability $\neq$ Impossible

- Suppose we have a continuous sample space. $\Omega = \{x : x \in [a, b]\}$
- We could think of human height, the failure rate of a government, or the length of time it takes to get from Mandalay to Bagan on a riverboat.
- Technically, no individual point in a continuous sample space can have positive probability.

Example

- Buses can range from being 0 to 10 minutes late. E might be the probability that a bus is more than 8 minutes late. Suppose that bus times are uniformly distributed:

$$P(\text{Late}) = \int_8^{10} f_X(x) dx$$

$$P(\text{Late}) = \int_8^{10} \frac{1}{10} dx$$

$$P(\text{Late}) = \frac{10}{10} - \frac{8}{10} = \frac{2}{10}$$

Summary of Terms/Concepts

- Sample Space, Event, Probability Law
- 3 Axioms of Probability
- Mutually Exclusive Events
- Uncountable Sample Spaces